To measure monotonicity properties of the function f_{LLM} we use Spearman rank correlation that we briefly describe next. Let $X, Y \in \mathbb{R}^n$ be two real n-dimensional vectors and let $R(X)$ (resp. $R(Y)$) denote an n-dimensional vector obtained from X (resp. Y) where the entries are substituted with their ranks in the sequence of sorted entries of X (resp. Y). Then, Spearman rank correlation coefficients (usually denoted by ρ) is given by:

$$\rho = \frac{\mathbf{Cov}(R(X), R(Y))}{\sigma(R(x)) \cdot \sigma(R(Y))}, \tag{3}$$

where $\mathbf{Cov}(R(X), R(Y))$ is the covariance between rank vectors $R(X)$ and $R(Y)$, while $\sigma(R(X))$ and $\sigma(R(Y))$ are their respective standard deviations.

Spearman coefficient ρ is a nonparametric measure for the alignment of the two vectors. Loosely speaking, the coefficient assesses if the increment in one variable corresponds to the increase (or decrease) of the other. In particular, unlike Pearson coefficient which takes into account the value of the changes, Spearman's ρ takes into account only the sign of the changes.

3.4 Scaling Rate Index

For a monotonically increasing sequence of positive real numbers $x_1, \ldots, x_n$, we introduce the Scaling Rate Index to measure the rate at which numbers grow in magnitude. More precisely, we seek for positive real constants α and β that minimize the following objective function:

$$\sum_{i=1}^{n-1} |(x_{i+1} - x_i) - \alpha \cdot \beta^i|^2. \tag{4}$$

In particular, if $\beta > 1$, the difference between two consecutive members x_i and x_{i+1} is expected to increase with i. In other words, $x_{i+1} - x_i$ is expected to be smaller than $x_{i+2} - x_{i+1}$. Such sequences are commonly known as *convex sequences* (Rockafellar, 2015) and the growth of x_i is exponential (superlinear) in i. An example of such a sequence that is relevant to our study is that defined with $x_i := 10^i$. Then, $x_{i+1} - x_i = 9 \cdot 10^i$ and we can take $\alpha = 9$ and $\beta = 10$.

If $\beta = 1$, expression (4) indicates that the difference between two consecutive members x_i and x_{i+1} is approximately constant, i.e. the sequence is approximately linearly increasing. For example, one may take the sequence $x_i := i$, for which $\alpha = 1 = \beta$.

Finally, $\beta < 1$ implies that the difference between consecutive members is steadily decreasing, $x_{i+1} - x_i$ is expected to be greater than $x_{i+2} - x_{i+1}$. Such sequences are commonly known as *concave sequences* (Rockafellar, 2015), and the growth of the sequence is exponentially decreasing (sublinear). An example of such a sequence is given by $x_i = 1 - \frac{1}{10^i}$, with $\alpha = 9$ and $\beta = \frac{1}{10}$.

4 Experiment 1: Identifying number line using contextualized numbers

Setup. To systematically probe the model's numerical representations, we partition numbers into logarithmically spaced groups:

$$\begin{aligned} &G_1 = \{1, 2, \ldots, 20\}, \\ &G_{i+1} = \{10^i - 19, \ldots, 10^i + 20\}, \quad i \geq 2. \end{aligned} \tag{5}$$

These groups ensure that larger group indices correspond to numerical magnitudes that increase exponentially with index, reflecting the logarithmic nature of the mental number line hypothesis.

To analyze the embeddings of numbers, to every number $x \in G_i$ we assign the following prompt:

$$x \leftarrow \texttt{a=a, b=b, c=c, x=} \tag{6}$$

where `a, b, and c` are randomly generated numbers from the groups G_i. This prompt structure is designed to provide the model with contextual examples, encouraging it to invoke the number x in model's hidden states representations (see Figure 2). Such approaches have been used in prior work to probe contextual representations in language models Srivastava et al. (2024).

The hidden state representation $f(x) \in \mathbb{R}^d$ is extracted from a designated layer of the model from the last token in the prompt, e.g. the '=' token. We use only those x for which the generated output of the model is x itself.

To ensure a representative sampling, we randomly select k numbers from each group G_i. These sampled numbers collectively form a dataset, denoted as $\mathcal{X}$, which serves as the basis for our analysis. The set of hidden state representations, $\{f(x)\}_{x \in \mathcal{X}}$, is then aggregated and analyzed to investigate patterns and properties in the embedding space.

To control for potential biases introduced by tokenization (where larger numbers often span more tokens) we conduct a complementary experiment

using non-numerical sequences. Instead of numerical inputs, we construct sequences of random letters with lengths corresponding to the tokenized representations of numbers. The letter sequences are grouped to their lengths so that the grouping approximately matches one of the numbers, and the prompts corresponding to specific letter sequences are designed in a similar fashion as for the numbers (6). This setting allows us to compare any observed structural patterns between the number representations and letter representations. By doing so, we can determine whether the model truly encodes numerical magnitude or if it is simply responding to surface-level features of the input.

Motivation. The goal of this experiment is twofold. We first investigate whether LLMs encode numerical values in context along a *monotonic number line* in their internal representation space. Second, we test whether this number line exhibits *sublinear scaling*, similar to human cognitive representations of numbers.

Methodology. For these purposes, we use:

• PCA nad PLS. After the set of hidden state representations is aggregated, we further project it into a one-dimensional space using PCA or PLS methods. In particular, for PLS we take the numbers themselves to form the target vectors, while for letters, we consider the letter sequence as a base 26 representation of a number (with random but fixed assignment of values to the letters), and use this number as the corresponding target.

• Monotonicity metric. We apply it on a sequence $x_1, \ldots, x_n$ of all the numbers in the union of groups G_j defined in (5), and their respective projections $f_{\text{LLM}}(x_1), \ldots, f_{\text{LLM}}(x_n)$. Spearman rank coefficient will tell us whether the model preserves natural ordering of the numbers.

• Scaling Rate Index. The initial centers of G_i, given by $x_i = 10^i$, form a convex, exponentially growing sequence characterized by parameters $\alpha = 9$ and $\beta = 10$. These numbers serve as representative scales of the numbers within each group.

To obtain a robust estimate of how these scales are preserved in the projections under f_{LLM}, we compute the expectation of projections in each group. Specifically, for SRI analysis, we define the sequence

$$\bar{f}_{\text{LLM}}(i) = \mathop{\mathbb{E}}_{x \in G_i} [f_{\text{LLM}}(x)], \tag{7}$$

and fit positive constants α and β such that $\bar{f}_{\text{LLM}}(i) \approx \alpha \cdot \beta^i$.

In particular, the mapping

$$10^i \mapsto \bar{f}_{\text{LLM}}(i) \tag{8}$$

allows us to examine how does f_{LLM} scales numerical magnitudes. The fundamental question we seek to answer is: **What is the nature of the function f_{LLM} (logarithmic, linear, or exponential)**?

To answer this question, we analyze the scaling factor β in the fitted exponential model[4]. In the following, we explain how different values of β correspond to the underlying properties of f_{LLM}.

• If $\beta > 1$, the sequence $\bar{f}_{\text{LLM}}(i)$ is convex and exponentially increasing. This means f_{LLM} maps an exponentially increasing sequence to another exponentially increasing sequence:

$$9 \cdot 10^i \mapsto \alpha \cdot \beta^i = \alpha \cdot 10^{(\log_{10} \beta) \cdot i}.$$

Thus, f_{LLM} preserves the original spacing of numbers, albeit with a scaling factor $\log_{10} \beta > 0$.

• If $\beta = 1$, the sequence $\bar{f}_{\text{LLM}}(i)$ is linearly increasing, meaning f_{LLM} takes the form:

$$10^i \mapsto \alpha \cdot i = \alpha \cdot \log_{10} 10^i.$$

In this case, f_{LLM} exhibits logarithmic scaling.

• If $\beta < 1$, the sequence $\bar{f}_{\text{LLM}}(i)$ is concave, exponentially decaying. Here, f_{LLM} follows:

$$10^i \mapsto \alpha \cdot \beta^i = \alpha \cdot 10^{-(\log_{10} \frac{1}{\beta}) \cdot i}.$$

Thus f_{LLM} is a *sub-logarithmic* mapping.

Results. The results reveal distinct yet consistent patterns in how different models encode numerical and alphabetical structures, with variations across layers (Tables 1 and 2). Despite these variations, similar trends emerge across the models, leading to consistent conclusions about their processing of numerical values (please refer to appendix A for experimental details).

Numerical vs. Symbolic Representations. First key finding is that numerical embeddings exhibit a significantly higher explained variance

[4]We can disregard α from the analysis since it does not influence the scaling but merely introduces a bias.

Model	Group	Layer	$\rho \pm$ std	$\beta \pm$ std	$\sigma^2 \pm$ std
LLaMA-2-7B	Numbers	3	0.97 ± 0.00	0.83 ± 0.06	0.60 ± 0.01
	Letters	1	0.45 ± 0.00	1.21 ± 0.00	0.24 ± 0.00
Pythia-2.8B	Numbers	8	0.94 ± 0.01	0.54 ± 0.01	0.31 ± 0.01
	Letters	11	0.89 ± 0.01	0.53 ± 0.10	0.16 ± 0.01
GPT-2-L	Numbers	18	0.95 ± 0.00	0.58 ± 0.02	0.32 ± 0.00
	Letters	5	0.11 ± 0.05	0.80 ± 0.42	0.21 ± 0.01
Mistral-7B	Numbers	3	0.96 ± 0.00	1.05 ± 0.00	0.44 ± 0.00
	Letters	14	0.89 ± 0.00	0.60 ± 0.00	0.22 ± 0.00
LLaMA-3.1-8B	Numbers	1	0.41 ± 0.04	1.14 ± 0.05	0.48 ± 0.01
	Letters	1	0.56 ± 0.00	0.16 ± 0.07	0.19 ± 0.01
LLaMA-3.2-Instruct-1B	Numbers	4	0.93 ± 0.02	1.33 ± 0.12	0.35 ± 0.01
	Letters	1	0.57 ± 0.06	0.47 ± 0.08	0.17 ± 0.00

Table 1: Comparison of several models on Numbers and Letters groups, evaluated using three metrics: ρ, β, and Explained Variance (σ^2). Results are reported for the layer with the highest σ^2 score. Standard deviations are included.

Model	Group	Layer	$\rho \pm$ std	$\beta \pm$ std	$R^2 \pm$ std
Llama-3.2-1B-Instruct	Numbers	6	0.91 ± 0.00	1.93 ± 0.05	0.68 ± 0.01
	Letters	10	0.93 ± 0.00	0.97 ± 0.03	0.45 ± 0.03
Pythia-2.8b	Numbers	1	0.78 ± 0.02	4.65 ± 1.32	0.71 ± 0.01
	Letters	20	0.90 ± 0.01	0.95 ± 0.11	0.46 ± 0.04
GPT2-L	Numbers	17	0.96 ± 0.01	1.15 ± 0.09	0.67 ± 0.03
	Letters	33	0.81 ± 0.03	0.93 ± 0.04	0.44 ± 0.01
Llama-2-7b	Numbers	5	0.93 ± 0.00	2.62 ± 0.00	0.81 ± 0.00
	Letters	27	0.88 ± 0.03	0.91 ± 0.02	0.45 ± 0.01
Mistral-7B-v0.1	Numbers	7	0.88 ± 0.00	14.87 ± 8.28	0.81 ± 0.00
	Letters	29	0.86 ± 0.00	1.62 ± 0.00	0.63 ± 0.00
Llama-3.1-8B	Numbers	4	0.93 ± 0.01	2.00 ± 0.01	0.73 ± 0.01
	Letters	16	0.93 ± 0.01	0.88 ± 0.06	0.45 ± 0.02

Table 2: Comparison of several models on Numbers and Letters groups, evaluated using three metrics: ρ, β, and R^2. Results are reported for the layer with the highest R^2. Standard deviations are included.

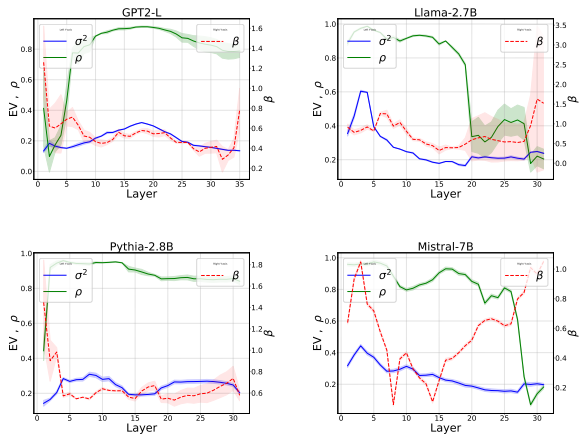

Figure 3: Layer-wise analysis of four models on numerical groups, showing explained variance (σ^2), monotonicity (ρ), and Scaling Rate Index (β). The layer with maximum σ^2 aligns with peak ρ, indicating optimal numerical encoding.

(σ^2 in Table 1 and R^2 in Table 2) in the one-dimensional PCA and PLS transformations compared to letter-based embeddings (refer to Methodology). This suggests that numbers naturally align along a one-dimensional manifold—akin to a number line—while random sequences of letters do not display the same structured behavior.

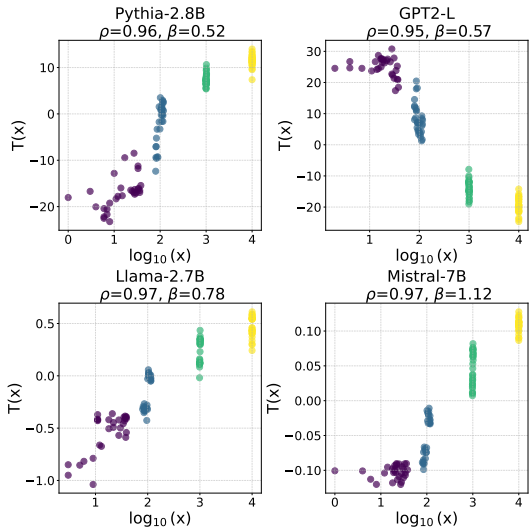

Figure 4: Projections of numerical representations (y-axis) against their log-scaled magnitudes (x-axis) for the layer with the highest explained variance in four models. Sublinearity and monotonicity (ρ) are indicated above each subfigure, demonstrating consistent sublinear trends and strong monotonic relationships across models.

Furthermore, the monotonicity metric (ρ) consistently shows higher values for numerical data, with most models achieving $\rho > 0.9$ in both PCA and PLS analyses. This supports the idea that numerical representations are not only structured, but also maintain a well-ordered progression across layers.

The resulting projections obtained using PCA for the numerical and letters groups are visualized in Figures 4 and 5, respectively.

Sublinearity in Numerical Representations. The sublinearity coefficient (β) derived from PCA projections reveals notable differences across models. Some, such as LLaMA-2-7B, Pythia, and GPT-2 Large, exhibit strong sublinear (sublogarithmic) scaling with $\beta < 1$, indicating that embedding distances grow at a diminishing rate. In contrast, models like Mistral show a nearly logarithmic trend ($\beta \approx 1$), while others approach a more linear spacing pattern with higher β values.

Layer-Wise Dynamics. Since Table 1 reports values from the layer with the highest explained variance, interpretation requires caution—other lay-